

B 2.5.1 File Processing



B 2.5.1 Construct code to perform file-processing operations.

Learn file processing in Java and Python. This resource covers opening, reading, writing, and appending to text files, including error handling and best practices. Hands-on activities, worksheets, and slide decks for both languages provide practical coding experience. Students develop critical thinking and communication skills.

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Learning Objectives

- Students will understand the need for file processing in computer programs.
- Students can open, read, write, and append to text files in code.
- Students will be able to handle potential exceptions during file operations.
- Students will understand the importance of closing files to prevent data corruption.
- Students will be able to choose appropriate file-processing methods for different tasks.

IB ATL Skills

Thinking: Critical thinking (analysing file processing scenarios, choosing appropriate methods)

Communication: Explaining file processing code and its functionality.

Self-Management: Time management during coding activities.

Research: Investigating different file processing libraries/methods.

Lesson Plan

Introduction (15 minutes)

Activity: Brainstorming: Ask students about situations where programs need to interact with files (e.g., saving game progress, reading configuration files, processing data). Discuss the limitations of storing data only in variables.

File Processing Concepts (20 minutes)

Presentation: Introduce the concepts of file paths, file modes (read, write, append), and the steps involved in file processing (open, read/write, close).

Discussion: Emphasise the importance of error handling (e.g., file not found) and closing files.

Coding Demonstration (30 minutes)

Java Slide Deck: Demonstrate file operations using Scanner, FileWriter, and BufferedReader.

Python Slide Deck: Demonstrate file operations using open(), read(), readline(), write(), and close().

Explanation: Explain each line of code clearly, highlighting the syntax and the purpose of each operation.

Guided Practice (40 minutes)

Activity: Provide students with coding exercises (Worksheet 1) that covers the basics of the learning statements.

Support: Circulate around the classroom, providing assistance and answering questions.

Independent Practice (15 minutes)

Activity: Assign a more challenging task (Worksheet 2) that requires students to apply their knowledge of file processing to solve a problem. Example: A program to analyse a text file and count word occurrences.

Lesson Resources

Presentations

Java

[Link to presentation](#)

Python

[Link to presentation](#)

eBook

Two eBook pages for students provide clear, step-by-step guides to working with files. These pages can be used for reading tasks or even presented in lessons (in presentation mode).

B 2.5.1 Java

B 2.5.1 Python

Worksheets

Two worksheets with practical programming challenges will help students develop their understanding of file handling and their programming and problem-solving skills. The answer sheets include possible approaches.

B 2.5.1 Worksheet Java

B 2.5.1 Worksheet Java Answers

B 2.5.1 Worksheet 1 Python

B 2.5.1 Worksheet 1 Python Answers

B 2.5.1 Worksheet 2 Python

B 2.5.1 Worksheet 2 Python Answers

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